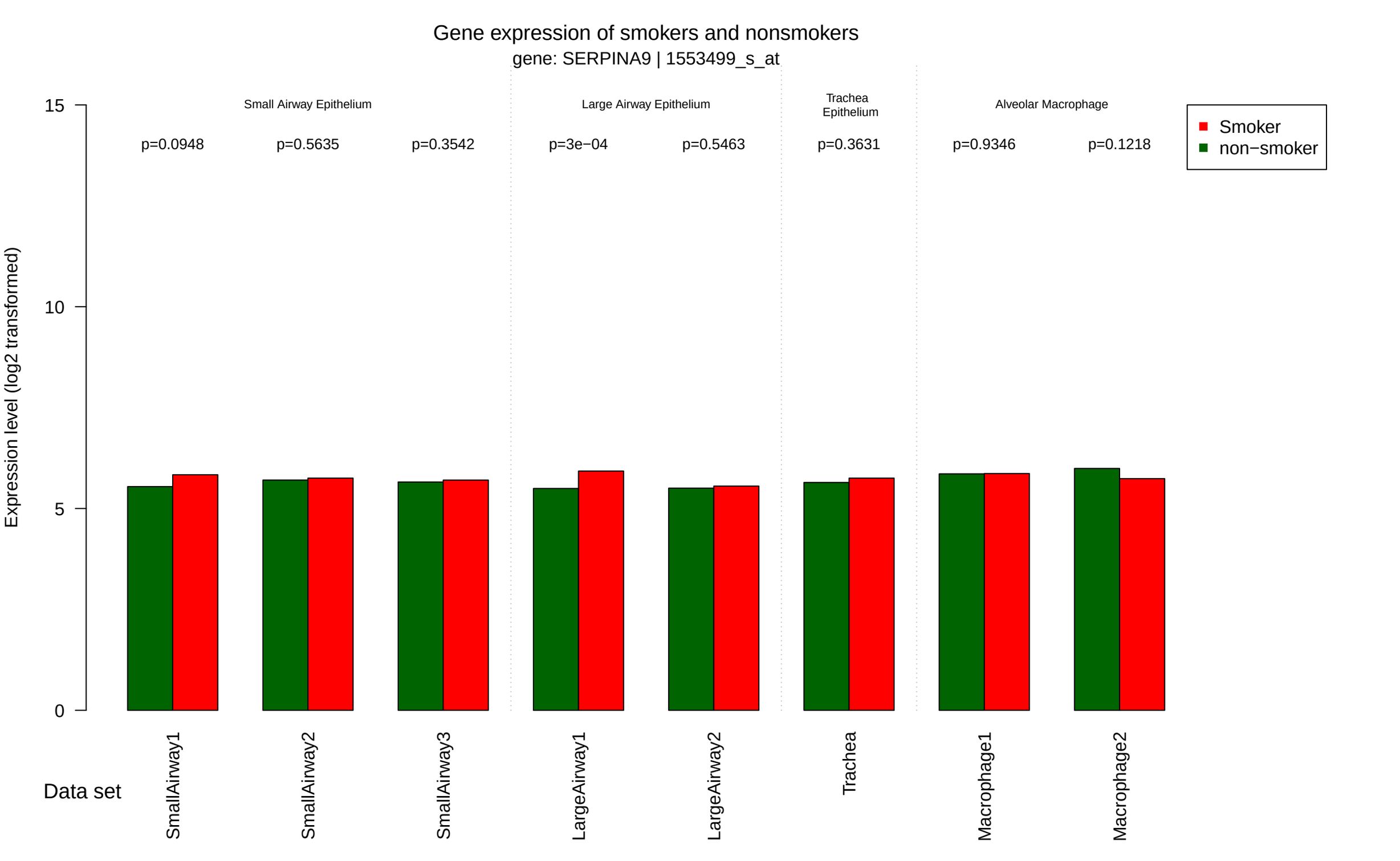
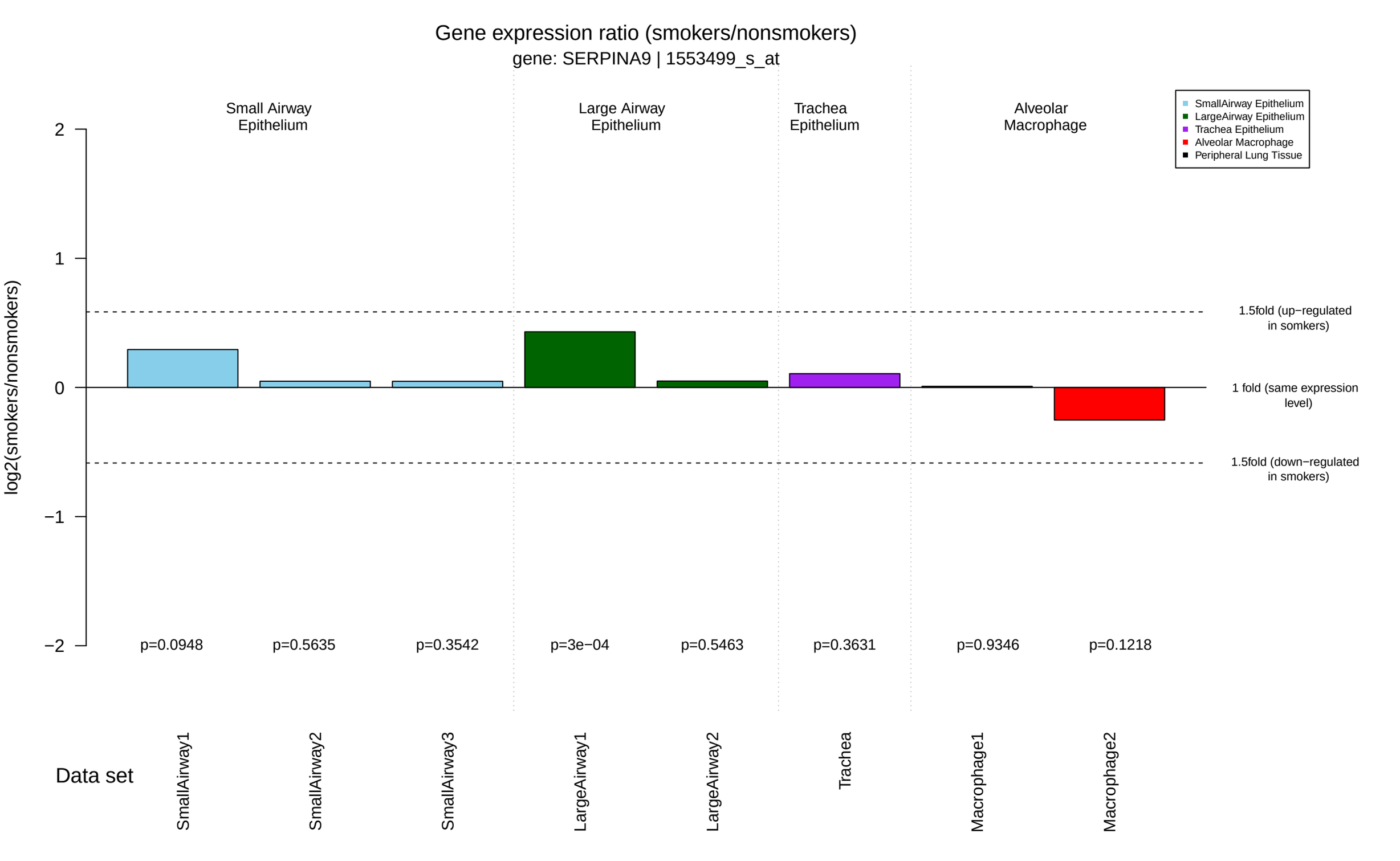
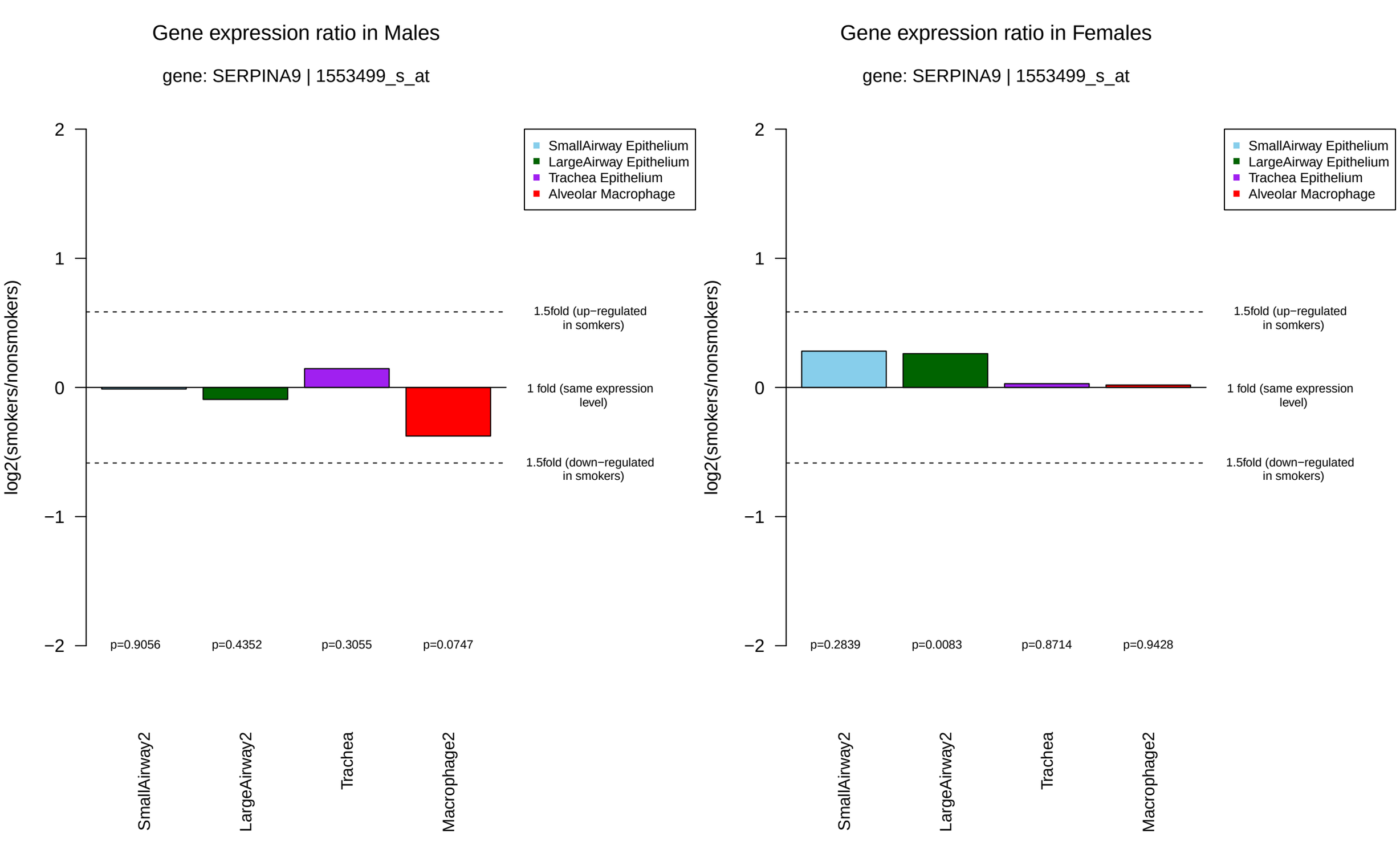
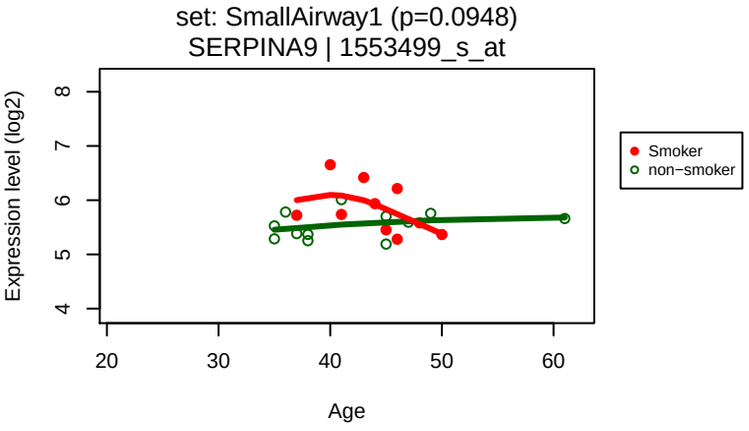
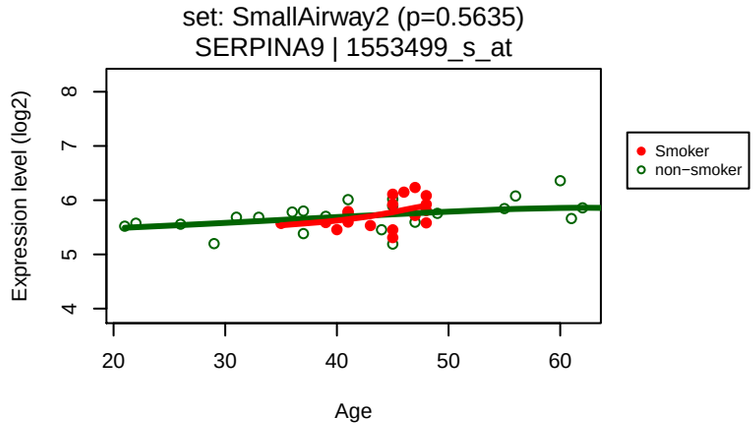
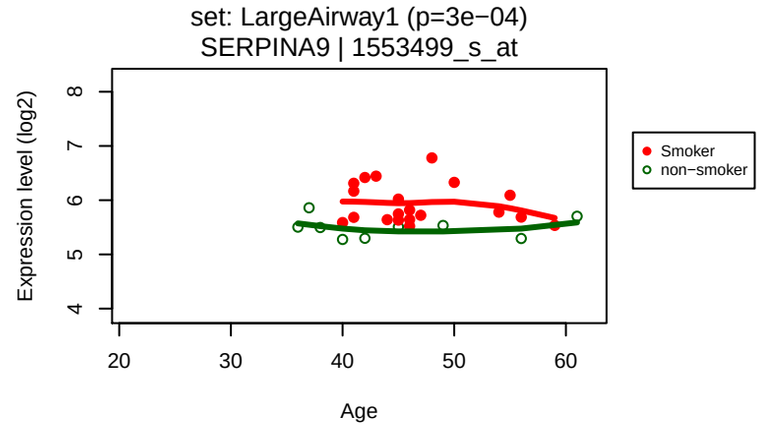

Gene expression of smokers and nonsmokers
gene: SERPINA9 | 1553499_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0948
p=0.5635
p=0.3542
p=3e−04
p=0.5463
p=0.3631
p=0.9346
p=0.1218
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

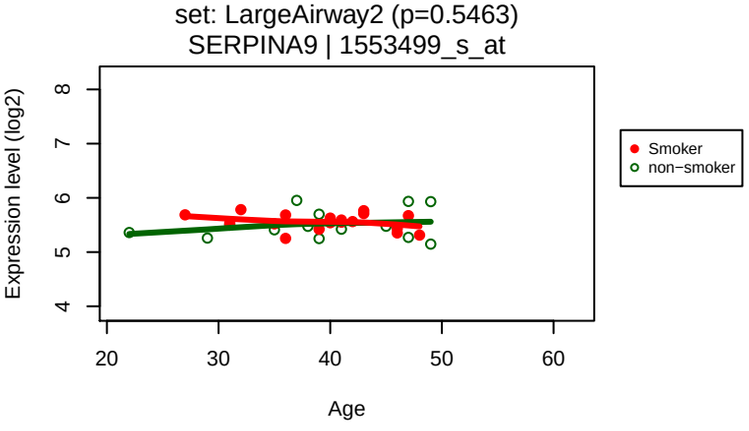

Gene expression ratio (smokers/nonsmokers)
gene: SERPINA9 | 1553499_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0948
p=0.5635
p=0.3542
p=3e−04
p=0.5463
p=0.3631
p=0.9346
p=0.1218
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

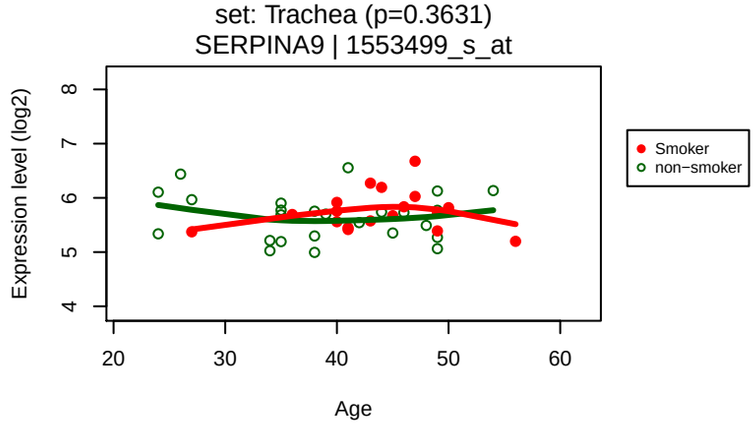

Gene expression ratio in Males
gene: SERPINA9 | 1553499_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9056
p=0.4352
p=0.3055
p=0.0747
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERPINA9 | 1553499_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2839
p=0.0083
p=0.8714
p=0.9428
SmallAirway2
LargeAirway2
Trachea
Macrophage2

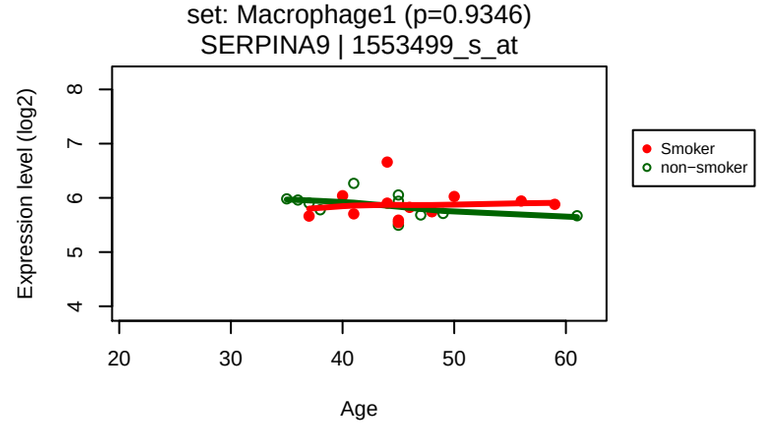

set: SmallAirway1 (p=0.0948)
SERPINA9 | 1553499_s_at
Expression level (log2)
Age
Smoker
non−smoker

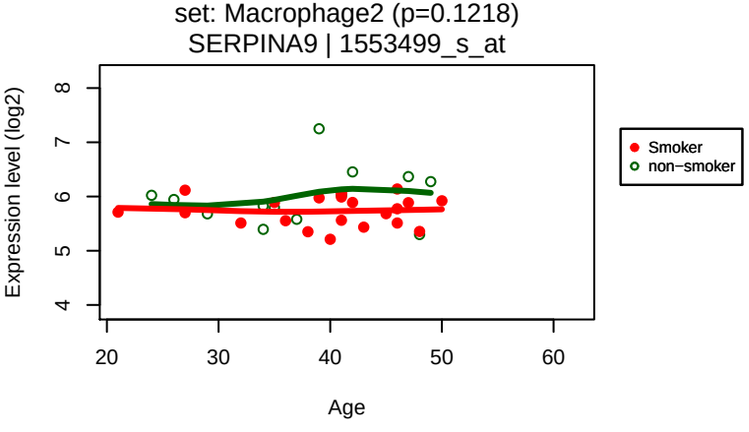

set: SmallAirway2 (p=0.5635)
SERPINA9 | 1553499_s_at
Expression level (log2)
Age
Smoker
non−smoker

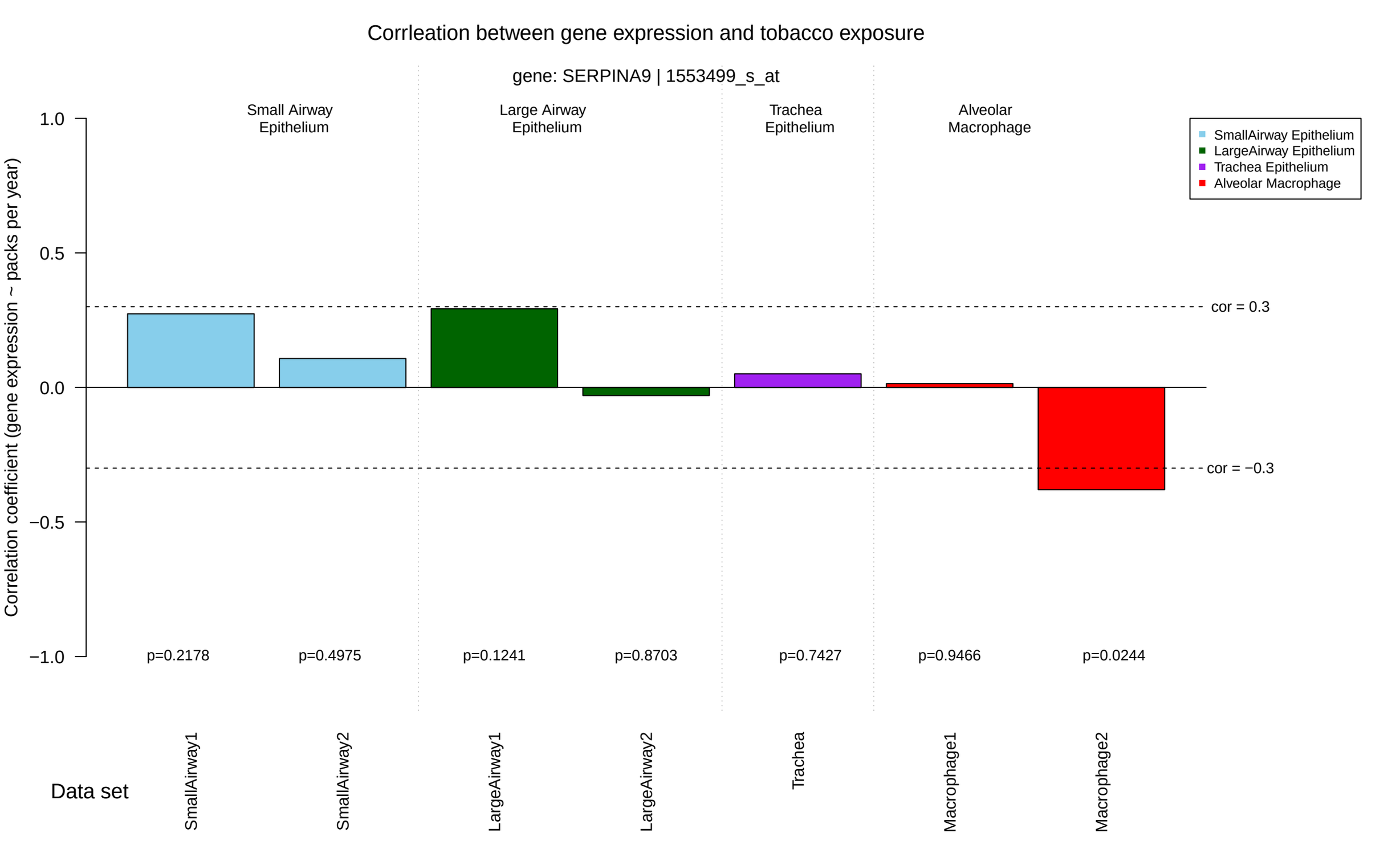

set: LargeAirway1 (p=3e−04)
SERPINA9 | 1553499_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5463)
SERPINA9 | 1553499_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3631)
SERPINA9 | 1553499_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9346)
SERPINA9 | 1553499_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1218)
SERPINA9 | 1553499_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERPINA9 | 1553499_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2178
p=0.4975
p=0.1241
p=0.8703
p=0.7427
p=0.9466
p=0.0244
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2